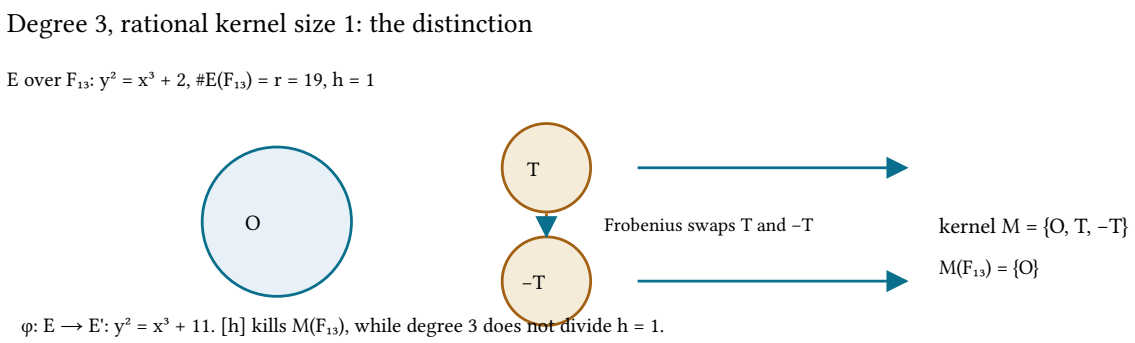
Isogeny and torsion fibers: a corrected cofactor statement

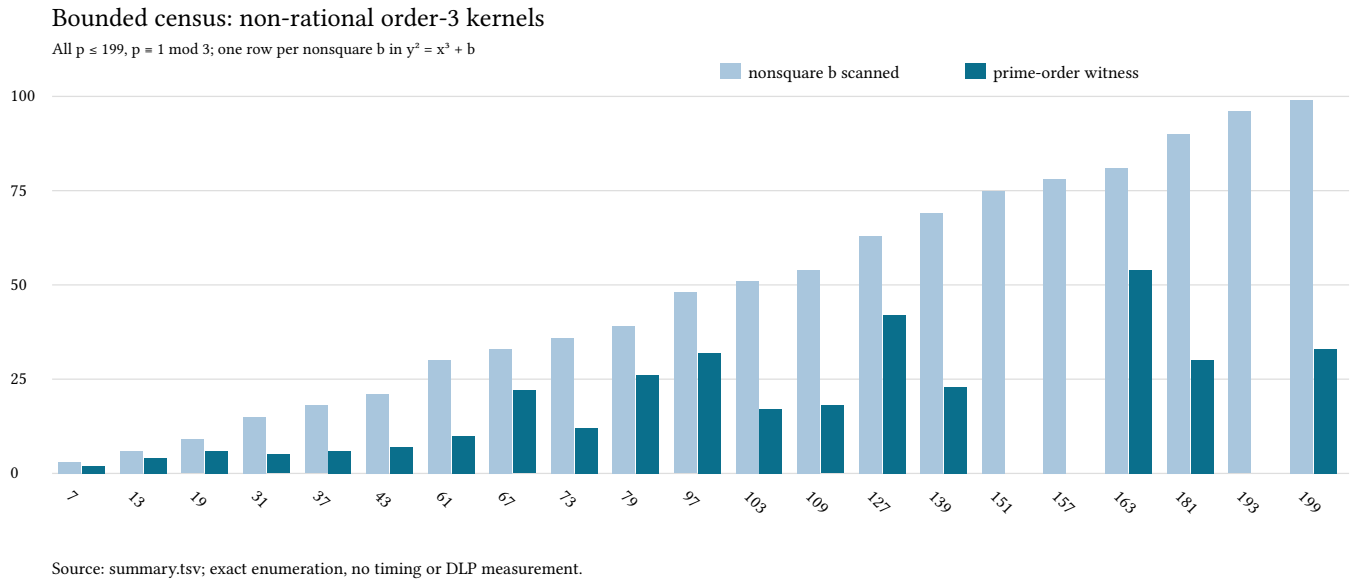
2026-10-12 · Structural correction · T-2070

The geometric kernel of a separable isogeny has size equal to its degree. Only the rational kernel controls rational fibers. The previous fiber report said the full degree d divides the cofactor h when the target prime r does not divide d. That implication is false. The corrected statement is: if K is the rational kernel and the curve order is h r, then |K| divides h.

Proof: |K| divides h r by Lagrange and divides d by the kernel order. Since r is prime and does not divide d, |K| is coprime to r. Euclid's lemma gives |K| divides h. Hence cofactor multiplication kills every rational kernel ambiguity in a lifted relation. The arithmetic step is checked by `lean/FiberCofactor.lean`; the geometric bridges are explicit hypotheses of that Lean declaration and remain to be formalized.



On E/F_{13}: $y^2 = x^3 + 2$, the curve order is 19 and h = 1. A Frobenius-stable order-3 kernel at x = 0 has only the identity over F_{13}. The standard quotient formula maps to E': $y^2 = x^3 + 11$. A native check verified all 19 images and every one of 361 rational ordered-pair sums. This is an exact bounded control; full geometric formalization remains open.



Source: summary.tsv; exact enumeration, no timing or DLP measurement.

The frozen scan found 349 prime-order witnesses among 1,014 curves at 21 base primes. Two independent native point counts agreed for every curve. These are selected-family counts, not a density estimate or DLP cost.

For a speed question, a destination coordinate solver, transport costs, full relation and algebra costs, target recovery, and matched rho still need a same-target comparison. The next experiment is recorded in `problems/nonrational-isogeny-fiber-dlp-cost-gap.md`. Prior work on small rational torsion and isogeny-class transfer remains separate: Faugère et al. (EUROCRYPT 2014); Galbraith, Hess and Smart (EUROCRYPT 2002); Voloch, "The discrete logarithm problem on elliptic curves and descents." Exact URLs and source links are in `REPORT.md`. The full release-library test is still unverified under shared-host contention; see `validation/README.md` for the retained attempts.